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Identification and Control of Novel Growth Inhibitors in Fed-Batch Cultures of Chinese Hamster Ovary Cells

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ABSTRACT: Chinese hamster ovary (CHO) cells in culture are known to consume large amounts of nutrients and divert most of them toward byproducts, some of which, including lactate and ammonia, are known to be toxic in nature. Glucose limitation strategies can successfully control lactate accumulation in fed-batch cultures yielding higher peak cell densities and titers. Interestingly, even in such optimized cultures, cell growth slows and eventually stops, indicating the emergence of other factors that negatively affect cell growth. In this study, we employed omics techniques to identify and quantify nine compounds that are intermediates or byproducts of amino acid metabolism, and accumulate in fed-batch cultures. Treatment with these compounds either individually or in a combined fashion resulted in partial or complete cell growth inhibition. Careful control of selected amino acid concentrations between one-half and one millimolar during the growth phase of fed-batch cultures reduced accumulation of the inhibitory metabolites and allowed for higher peak cell densities and increased productivity.

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KEYWORDS: amino acid metabolism; HiPDOG; growth inhibitors; CHO cells

Introduction

The global therapeutic protein industry has a current market value estimate of US $140bn (Walsh, 2014). The majority of new therapeutic proteins are produced using the industry standard host cell line, Chinese hamster ovary (CHO) cells, with a small number of new proteins still produced in mouse myeloma cell line hosts (NS0 or SP2/0), mostly to take advantage of the unique glycosylation properties of such hosts. Over decades of expression system development, the therapeutic protein productivities using CHO cells in fed-batch or hybrid bioreactor operational modes are now approaching 1 g/L/day.

CHO cells in fed-batch culture consume large amounts of nutrients including glucose and amino acids and divert most of them toward metabolic byproducts, some of which are cytotoxic. Classically, lactate and ammonia are known to be major byproducts. Sufficiently elevated levels of lactate have been shown to inhibit growth and productivity of various mammalian cells lines in culture (Cruz et al., 2000; Lao and Toth, 1997). Ammonia accumulation, on the other hand, not only hurts growth and productivity but also has been shown to alter glycosylation of the recombinant protein being produced (Cruz et al., 2000; Hansen and Emborg, 1994; Kurano et al., 1990; Lao and Toth, 1997; Ozturk et al., 1992; Yang and Butler, 2000). The proliferative phase of cells in fed-batch cultures is almost always associated with concurrent lactic acid production (Frame and Hu, 1991; Mulukutla et al., 2010; Vander Heiden et al., 2009).

Over the past decade, various approaches have been employed to understand and control lactate metabolism of CHO cells (Kim and Lee, 2007; Ma et al., 2009; Mulukutla et al., 2012, 2015; Zhou et al., 2011). One method to control lactate metabolism, namely the HiPDOG (Hi-end pH delivery of glucose) strategy, involves the reduction of lactate accumulation through continuous monitoring and control of residual glucose concentrations at very low levels in the culture (Gagnon et al., 2011). Such control is accomplished by using pH as an indirect sensor of the metabolic state of cells and the residual bulk fluid glucose concentrations, and feeding glucose intermittently when residual glucose levels decrease to almost negligible levels. Using such a strategy, even at the industrial manufacturing scale, lactate accumulation can be very effectively suppressed. Such reduction in net lactate production generally yields higher peak cell densities and higher process productivity. Interestingly, although lactate is controlled at sub-growth inhibitory levels throughout the growth phase, cells still eventually stop dividing in these cultures, apparently indicating that other cell-growth inhibitors are accumulating to consequential levels.

In the present work, we identify and quantify novel cell growth inhibitors that accumulate in CHO cell fed-batch cultures that employ the HiPDOG strategy to control lactic acid production. Subsequently, we demonstrate that nutrient limitation methodologies can result in reduction in accumulation of these newly

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discovered inhibitors. Such reduced inhibitor accumulation in fed-batch cultures results in increased peak cell densities and productivities.

Materials and Methods

Cells, Medium, and 6-Well Plate Culture

CHO cell lines (A and B) utilizing a glutamine synthetase expression system (hereafter called GS-CHO) and expressing recombinant antibodies were used. Two types of proprietary medium were used: "Medium A" which is used for inoculation of the experiment (on Day 0), and "Medium B" which is the enriched nutrient solution used as a feed media for conventional and HiPDOG fed-batch processes (for more details, see Supplementary Materials). For investigating the effect of various novel inhibitors or the spent media components on cell growth, cultures were set up in 6-well plates. Purified compounds of the nine metabolites (Table I) were obtained from Sigma–Aldrich® (St. Louis, MO). Cells were inoculated at low viable cell densities ($0.05 - 0.1 \times 10^6$ cells/mL) in purified chemical supplemented fresh Medium A or combinations of fresh Medium A and spent medium (or spent medium filtrate/retentate). Triplicate cultures were setup for each condition in 6-well tissue culture plates with a working volume of 4 mL/well and cultivated for 4 to 6 days on a shaking platform (Orbital Shaker, Bellco Glass, Inc., Vineland, NJ) in a humidified incubator (Thermo Fisher Scientific, Waltham, MA) maintained at 36.5°C and 5% carbon dioxide levels.

Bioreactor Setup and Operation

For the first study (Figs. 1 and S1), two conditions were used: the conventional fed-batch process and the glucose restricted fed-batch process. In the glucose restricted fed batch process (hereafter called HiPDOG culture), glucose was limited using the HiPDOG technology (Gagnon et al., 2011). The pH dead-band used while HiPDOG control was operational was 7.025 ± 0.025. The production media used was Medium A and the feed media used was Medium B.

The conventional process was identical to the HiPDOG process with respect to inoculum cell density targeted (1×10^6 cells/mL), the media used, culture volume (1 L), the amount of feed added daily to the culture, and the process parameters including the temperature (36.5°C), pH (7.025 ± 0.025) and the agitation rate (267 rpm). The only difference was the glucose level. In the conventional culture, glucose was maintained at greater than 2 g/L between Days 2 and 5. Whereas, in the HiPDOG culture, the initial available glucose was consumed by the cells naturally until the glucose level fell to a point at which the cells began to also consume lactic acid (observed by a slight rise in pH of the culture), which commenced the HiPDOG feeding strategy. Post-day 5, the glucose levels in both the conditions were maintained at concentrations above 2 g/L by feeding glucose as necessary. Also, post-day 5, both cultures were treated similarly for the duration of the experiment. Viable cell density, lactate, and ammonia concentration in the cell culture medium were measured on a daily basis for both conditions using a Nova Bioprofile FLEX Analyzer (Nova Biomedical, Waltham, MA). Titer analysis was performed by protein A HPLC (model 1100 HPLC, Agilent Technologies, Inc., Santa Clara, CA, protein A column model 2-1001-00, Applied Biosystems, Foster City, CA). Spent media samples were collected on various days for amino acid and metabolomics analysis (see Supplementary Materials for details).

For bioreactor cultures that were part of the experiments investigating the effect of amino acid restriction (Fig. 6), the HiPDOG strategy was operational from the start of the culture until Day 7. Glucose was maintained above 2 g/L after Day 7. For cultures with control of four or eight amino acids, culture amino acid levels were measured on a daily basis and based on the consumption rates of each amino acid over the previous 24 h, the culture was supplemented with amino acid stock solutions such that the predicted amino acid levels at the next sampling point would be 0.5 mM. Independent of amino acid supplementation through stock solutions, feed medium (Medium B) was delivered through the HiPDOG scheme.

In the low amino acid processes designed to control the amino acid levels via a pre-determined feeding schedule without any manual amino acid supplementation (Fig. 7), the levels of the eight amino acids were adjusted in Medium B to higher levels based on the strategy discussed in the Results section. The modified Medium B is then used as the feed, which was delivered through the HiPDOG scheme. No additional amino acid supplementation was performed in these cultures.

Table I. Names and the metabolic sources of the nine metabolites identified as putative growth inhibitors accumulating in the HiPDOG culture.

Group	Metabolite	Metabolic source
1	3-(4-Hydroxyphenyl)lactate	Phenylalanine and tyrosine
	4-Hydroxyphenylpyruvate	
	Phenyllactate	Phenylalanine
2	Indole 3-lactate	Tryptophan
	Indole 3-carboxylate	
3	L-Homocysteine	Methionine
	2-Hydroxybutyrate	
4	Isovalerate	Leucine
5	Formate	Serine, threonine, and glycine

Results

Suppressing Lactate Production Using HiPDOG Strategy Facilitates Identification of Next Level of Cell Growth Inhibitors

Application of the HiPDOG strategy to control lactate production in fed-batch cultures of mammalian cells has been previously demonstrated (Gagnon et al., 2011). Figures 1 and S1 demonstrate that application of the HiPDOG technology to cell line A limits lactate production to much lower levels when compared to conventional fed-batch cultures. Suppression of lactic acid production in HiPDOG cultures also reduces the osmolality increase that normally occurs when in a pH-controlled bioreactor basic titrant is added to the culture to maintain pH (Fig. S1). Although, increased lactate and osmolality appear to be the two potential causes for the suppressed growth

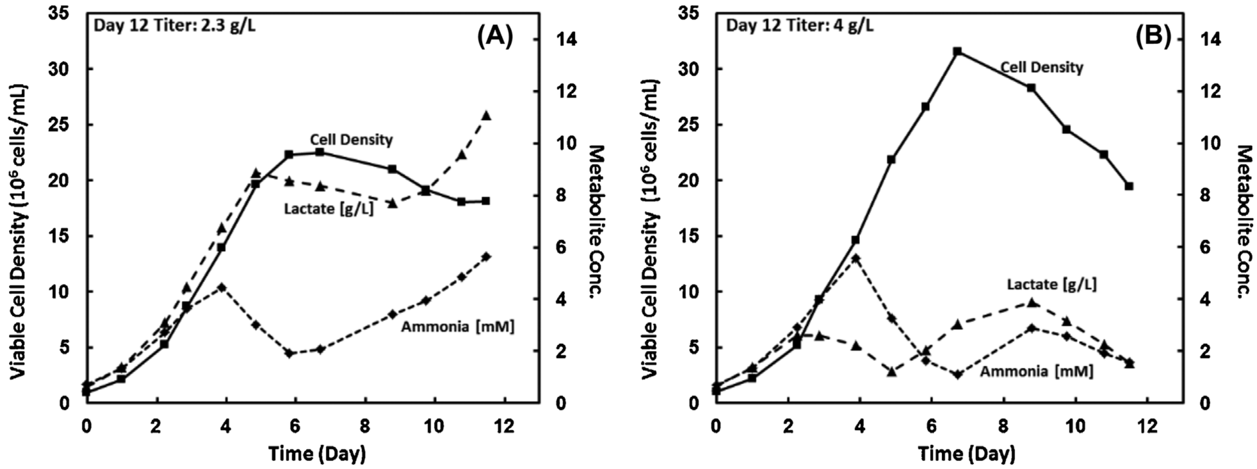

Figure 1. Growth and metabolic characteristics of CHO cell line A in fed-batch culture with or without employment of HiPDOG strategy. (**A**) Growth and metabolic phenotype of CHO cell line A in conventional fed-batch culture. (**B**) Growth and metabolic phenotype of CHO cell line A in fed-batch culture employing HiPDOG strategy. HiPDOG strategy was operational from Days 0 to 5. Plotted in each graph are the viable cell density (■), lactate (▲), and ammonia (◆) levels.

observed in the conventional fed-batch culture, more experimentation is needed to make a firm conclusion. However, reduced lactate accumulation and reduced osmolality increase seen in the HiPDOG culture seem to have translated into higher peak cell densities and higher productivities in the same when compared to conventional fed-batch cultures. In HiPDOG cultures, even though lactate, osmolality, and ammonia are controlled below their growth inhibitory levels, cell growth eventually slowed (as seen by cell density semi-log plot in Fig. S1), suggesting that other process related factors are coming into play once the production of classical inhibitors is controlled. The following factors were considered that might affect cell growth: process variables (temperature, pH, dissolved oxygen), biophysical factors (shear stress—agitation, bioreactor geometry), and biochemical factors (nutrient levels, osmolality, metabolite byproducts, host cell proteins). It was hypothesized that the factors contributing to cell growth inhibition were either metabolite byproducts (other than lactate or ammonia) or host cell proteins accumulating in the fed-batch cultures.

To determine whether accumulation of metabolite byproducts or host cell proteins was the predominant factor slowing cell growth, cells were removed on Day 7 from a bioreactor utilizing the HiPDOG lactate control technology, spun-down using a centrifuge, and re-suspended in various medium compositions at a low viable cell density and cultured in 6-well orbiting plates (see Materials and Methods section for more details). The following conditions were used: fresh medium identical to the initial fed-batch basal medium, spent medium from Day 7 of the HiPDOG bioreactor culture (a control sample was included to assess any effect of the spin-down and re-suspension procedure), and fresh medium with the addition of purified chemicals: sodium-L-lactate, ammonium chloride, or both (Fig. 2A). For conditions with add-back of purified chemicals, the substances were added to approximate the levels of lactate or ammonium that were measured in the HiPDOG bioreactor culture on Day 7 (when cell growth slows). A separate but similar experiment was performed to determine if host cell proteins or metabolic byproducts, other than lactate and ammonia, excreted, secreted, or merely expelled from lysed cells might slow cell growth (Fig. 2B). Day 7 cell-free culture supernatant from a sample taken from the HiPDOG bioreactor was ultra-filtered and concentrated 20-fold using an Ultracel 3 kDa (3000 Da) molecular weight (MW) cutoff centrifugal filter (Millipore) spun in a centrifuge (Sorvall® Legend® RT benchtop centrifuge) for 30–40 min at 3500 RPM. The filtrate (or the flow-through sample) should contain the compounds, metabolites, or proteins, with molecular weight below 3 kDa MW and the retentate (or the concentrated sample) should contain compounds mainly with molecular weight above 3 kDa MW. Cells were prepared in a similar manner to the previous experiment and were re-suspended in the following conditions: fresh medium identical to the initial fed-batch basal medium, fresh medium with serial additions of the spent bioreactor sample (Day 7) or the filtrate of the spent bioreactor sample (condition tests effect of low MW compounds), and fresh medium mixed with the retentate of the spent bioreactor sample (condition tests effect of high MW compounds). In the latter condition, the retentate of the spent media was mixed with the fresh medium such that the final level of the high MW compounds would be approximately twice more than those present in the original spent medium from Day 7 of the bioreactor run.

Cells were inoculated at low cell densities across these conditions and growth was monitored over the next 6 days. The flasks did not use any form of pH control (typically, pH dropped by 0.2 units in 5- or 6-day 6-well plate cultures) and at the cell densities used, depletion of any critical nutrient by end of the experiment was not expected. In the first experiment, cells grew well in the fresh medium condition and the fresh medium condition with lactate, ammonia, or both added back (Fig. 2A). Interestingly, significantly higher growth was observed in lactate supplemented conditions, the reasons for which are unclear (this phenomenon was reproducible; additional data not shown). Cell growth in the fresh medium condition of second experiment was

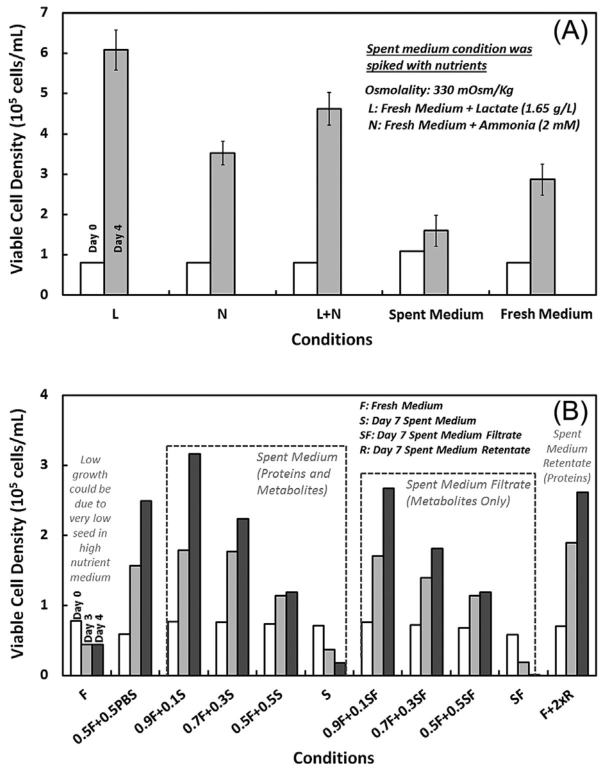

Figure 2. Effect of late stage HiPDOG fed-batch culture spent medium components on growth of CHO cell line A. (**A**) CHO cells from Day 7 of HiPDOG culture were spun down and resuspended in fresh media, fresh media supplemented with lactate (L) or ammonia (N) or both (L + N), or spent media from Day 7 of the HiPDOG culture. The lactate and ammonia concentration used were 1.65 g/L and 2 mM, respectively. (**B**) CHO cells from Day 7 of HiPDOG culture were spun down and resuspended in various conditions. The conditions include fresh medium (F), 50:50 mix of fresh medium and phosphate-buffered saline (0.5 F + 0.5 PBS), different combinations of fresh media (F) and Day 7 spent media (S) including 90:10 (0.9 F + 0.1 S); 70:30 (0.7 F + 0.3 S); 50:50 (0.5 F + 0.5 S); 100% spent media (S), different combinations of fresh media (F) and Day 7 spent media filtrate (SF) including 90:10 (0.9 F + 0.1 SF); 70:30 (0.7 F + 0.3 SF); 50:50 (0.5 F + 0.5 SF); 100% spent media (SF), or fresh media supplemented with spent media retentate such that the levels of compounds (high MW compounds) in the mix were at twice the levels as those in the spent media alone (F + 2 × R). Spent media filtrate is the flow through medium when spent medium is filtered using 3 KDa molecular weight cutoff filter. Spent media retentate is the concentrate obtained when the spent medium is filtered using 3 KDa molecular weight cutoff filter (see Materials and Methods section for experimental details).

hampered due to unknown reasons. High nutrient levels in the fresh medium along with the very low inoculum cell densities could be the reason for cell growth inhibition observed in fresh medium (Fig. 2B). Diluting the medium by 50% showed significantly higher growth. Interestingly, cells grew well in fresh medium with high molecular weight components added back (Fig. 2B). Cells in spent medium from Day 7 of HiPDOG culture exhibited slower growth (Fig. 2A and B). Interestingly, the cells in the conditions constituting serial dilutions of fresh medium with spent medium filtrate (or spent media) suffered growth proportional to the fraction of spent medium filtrate (or spent medium) used (Fig. 2B). Taken together, the data from the two experiments suggest that the small molecular weight compounds, potentially the metabolic byproducts or intermediates (other than lactate and ammonia), accumulating in the spent medium of the HiPDOG cultures by Day 7 are negatively affecting the cell growth. Host cell proteins above 3 kDa that accumulate in HiPDOG cultures by Day 7 seem to have no significant negative effect on cell growth.

Identification and Quantification of Metabolic Byproducts or Intermediates Accumulating in CHO Fed-Batch Cultures Using Metabolomics Analysis

A metabolomics approach was employed to profile the compounds accumulating in the HiPDOG cultures. Spent media samples from different time points of the HiPDOG culture were analyzed using LC-MS, GS-MS, and NMR. The metabolites that were accumulating to high levels on Day 7 were identified based on the fold changes measured. A fold change cut-off of 5, between the first time point the metabolite was detected, and the level measured on Day 7 of the HiPDOG culture, was employed to identify and classify a total of 18 compounds as those that accumulate to high levels. For each of the above-identified metabolites, the concentration at which it negatively affects cell growth was determined using spike-in experiments with purified compounds (see section, Add-back Experiments to Probe Inhibitory Concentrations). The results of these experiments were used to narrow the number of the putative novel inhibitors to 9. The list of the nine metabolites identified as potential inhibitors as well as their potential metabolic source in the cell culture medium are reported in Table I.

Using a LC-MS and GC-MS global metabolite profiling approach similar to that used for inhibitor identification and relative quantification, independent calibration curves for phenyllactate, 3-(4-hydroxyphenyl)lactate, 4-hydroxyphenylpyruvate, and indole 3-lactate were prepared. Figure S2 shows the calibration curve for phenyllactate as an example. These calibration curves are mathematical correlations of the actual amounts of the metabolite used in LC-MS or GS-MS techniques to the intensity values measured. The correlations were subsequently used along with the intensity values measured to calculate the concentration of each metabolite on Day 7 of the HiPDOG culture. Levels of indole 3-carboxylic acid on Day 7 were determined using a separate LC-MS method developed specifically for indole 3-carboxylic acid. The concentrations for the isovalerate and formate were determined by NMR technology as discussed in the Supplementary Materials section. The concentrations of these putative inhibitors on Day 7 of the HiPDOG culture are listed in Table II.

Table II. The concentrations of the putative inhibitors on Day 7 of the HiPDOG culture.

Serial #	Metabolite	Day 7 Concentration (mM)
1	3-(4-Hydroxyphenyl)lactate	0.38
2	4-Hydroxyphenylpyruvate	0.08
3	Phenyllactate	0.20
4	Indole 3-lactate	0.26
5	Indole 3-carboxylate	0.01
6	Isovalerate	2.41
7	Formate	3.97

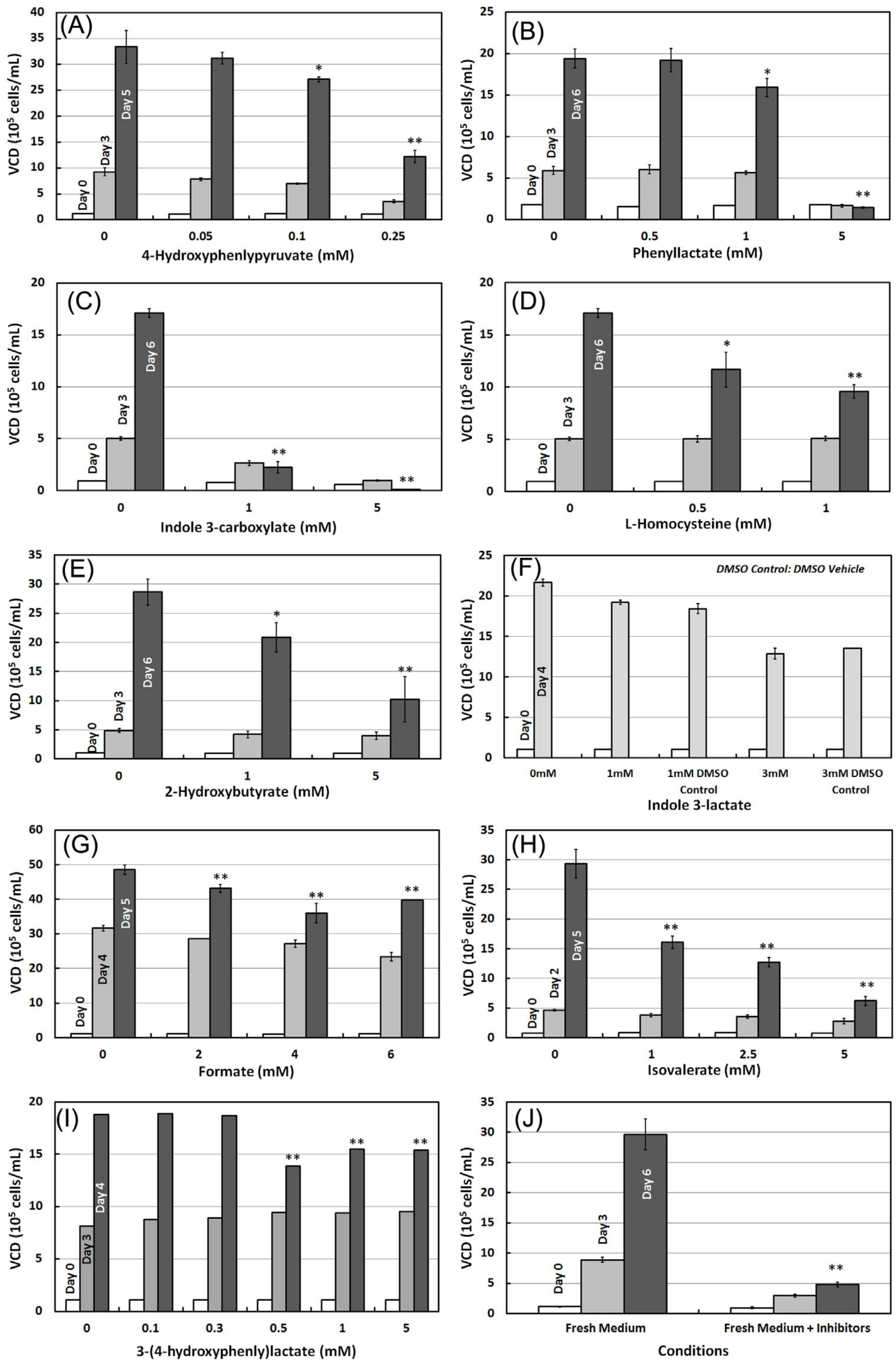

Figure 3. Effect of various metabolites accumulating in HiPDOG fed-batch cultures on growth of CHO cell line A. Effect of 4-Hydroxyphenylpyruvate (**A**), Phenyllactate (**B**), Indole 3-carboxylate (**C**), L-Homocysteine (**D**), 2-Hydroxybutyrate (**E**), Indole 3-lactate (**F**), Formate (**G**), Isovalerate (**H**), or 3-(4-hydroxyphenyl)lactate (**I**) on growth of CHO cell line A. In each case, freshly inoculated CHO cell cultures were treated with indicated concentrations of purified compounds, individually, and cell growth was monitored for 4 to 6 Days. (**J**) Synergistic effect of novel inhibitors on growth of CHO cell line A. CHO cells were inoculated either in fresh medium (Fresh Medium) or fresh medium supplemented inhibitors (Fresh Medium + Inhibitors) including phenyllactate, 4-hydroxyphenylpyruvate, 3-(4-hydroxyphenyl)lactate, isovalerate, formate, indole 3-carboxylate, and indole 3-lactate. Concentrations used for each compound are listed in Table II. Cell growth was monitored for 6 days post-treatment. Statistical significance of the difference in the Day 4 (or Day 5 or Day 6) viable cell densities between the treated and untreated conditions was assessed. * indicates $0.01 < P\text{-value} < 0.05$ and ** indicates $P\text{-value} < 0.01$. VCD, Viable Cell Density.

Add-Back Experiments to Assess the Inhibitory Concentrations of the Novel Growth Inhibitors

The independent effect of all nine inhibitors on growth of GS-CHO cells was investigated. Cells were inoculated at low cell densities in fresh medium and treated with different concentrations of various inhibitors. Growth kinetics were monitored for 4 to 6 days following treatment. Indole 3-carboxylate and 4-hydroxyphenylpyruvate were observed to have a potent negative effect on cell growth at concentrations of 1 mM or lower (Fig. 3A and C). Modest inhibition of growth was observed when GS-CHO cells were exposed to either homocysteine, 2-hydroxybutyrate, or 3-(4-hydroxyphenyl)lactate at concentrations higher than 0.5 or 1 mM (Fig. 3D, E, and I). Phenyllactate had a mild effect on cell growth at 1 mM concentration (Fig. 3B). Indole 3-lactate showed little or no effect at concentrations of 3 mM or below when compared to DMSO vehicle control (Fig. 3F). Formate had a negative effect on growth at 2 mM and beyond (Fig. 3G). Isovalerate had a significant negative effect on cell growth at concentrations of 1 mM and beyond (Fig. 3H).

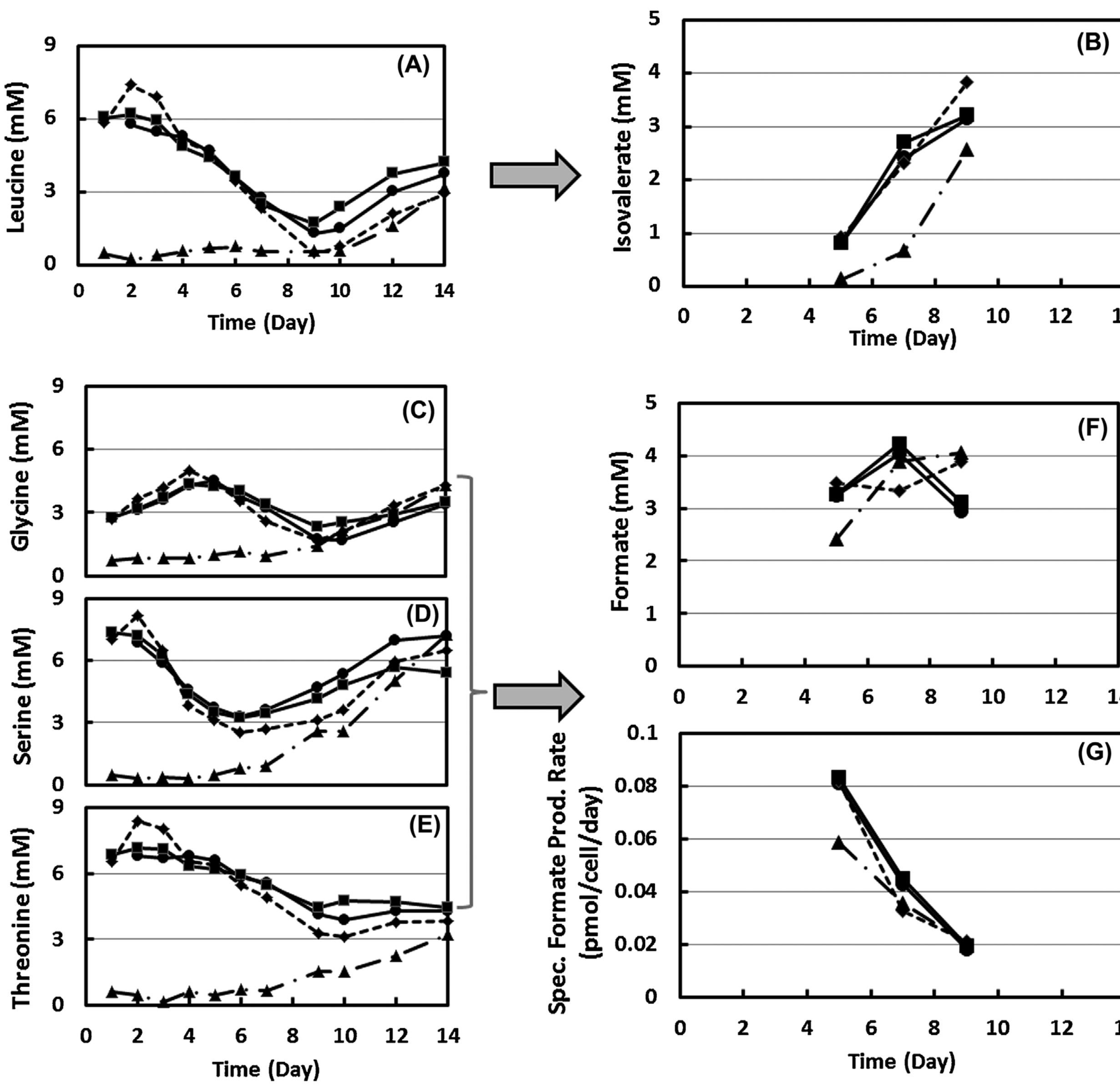

Figure 4. Effect of nutrient limitation on accumulation of the novel growth inhibitors in HiPDOG fed-batch cultures of CHO cell line A. HiPDOG cultures of CHO cells were carried out by maintaining the levels of eight amino acids (Low 8AA) or four amino acids (Low 4AA) in concentration range 0.5–1 mM for first 7 days of the culture. Control HiPDOG cultures (HiPDOG 1 and 2) were carried at normal amino acid levels. Leucine (**A**) was controlled at low levels only in the Low 8AA condition and led to lower accumulations of isovalerate (**B**) in culture. Glycine (**C**), serine (**D**), and threonine (**E**) were controlled at low levels only in the Low 8AA condition which resulted in lower (Day 5) or similar (Day 7) levels of formate (**F**) in the Low 8AA condition as compared to control cultures or the Low 4AA culture. The specific production rate of formate (**G**) was lower in the Low 8AA condition. Low 8AA amino acids: tyr, phe, met, trp, leu, ser, met, thr; Low 4AA: tyr, phe, met, trp. Low 8AA (▲), Low 4AA (◆), HiPDOG 1 (■), and HiPDOG 2 (●).

The concentration at which cell growth was affected by inhibitors individually in the add-back experiments was much higher than the concentration of the inhibitors measured on Day 7 of the HiPDOG fed-batch culture (Table II). Therefore, the effect of these inhibitors when treated in combination was subsequently investigated. Treatment of cells with the combination of just seven (phenyllactate, 4-hydroxyphenyllactate, 3-(4-hydroxyphenyl)lactate, indole 3-lactate, indole 3-carboxylate, isovalerate and formate) of the nine metabolites, at concentrations as measured on Day 7 of the HiPDOG culture, resulted in significant growth inhibition as compared to cell growth in fresh medium (Fig. 3J). These data indicate that the seven metabolites mentioned earlier act in a synergistic manner to inhibit cell growth. The above-mentioned nine metabolites are catabolic intermediates or by-products of phenylalanine, tyrosine, tryptophan, leucine, serine, threonine, methionine, or glycine metabolism. It was, therefore, hypothesized that suppressing the uptake rates of the above-mentioned amino acids might result in reduced biosynthesis of the corresponding inhibitors. Such reduction in amino acid uptake rates can be accomplished by restricting the residual levels of amino acids in fed-batch cultures to lower concentrations.

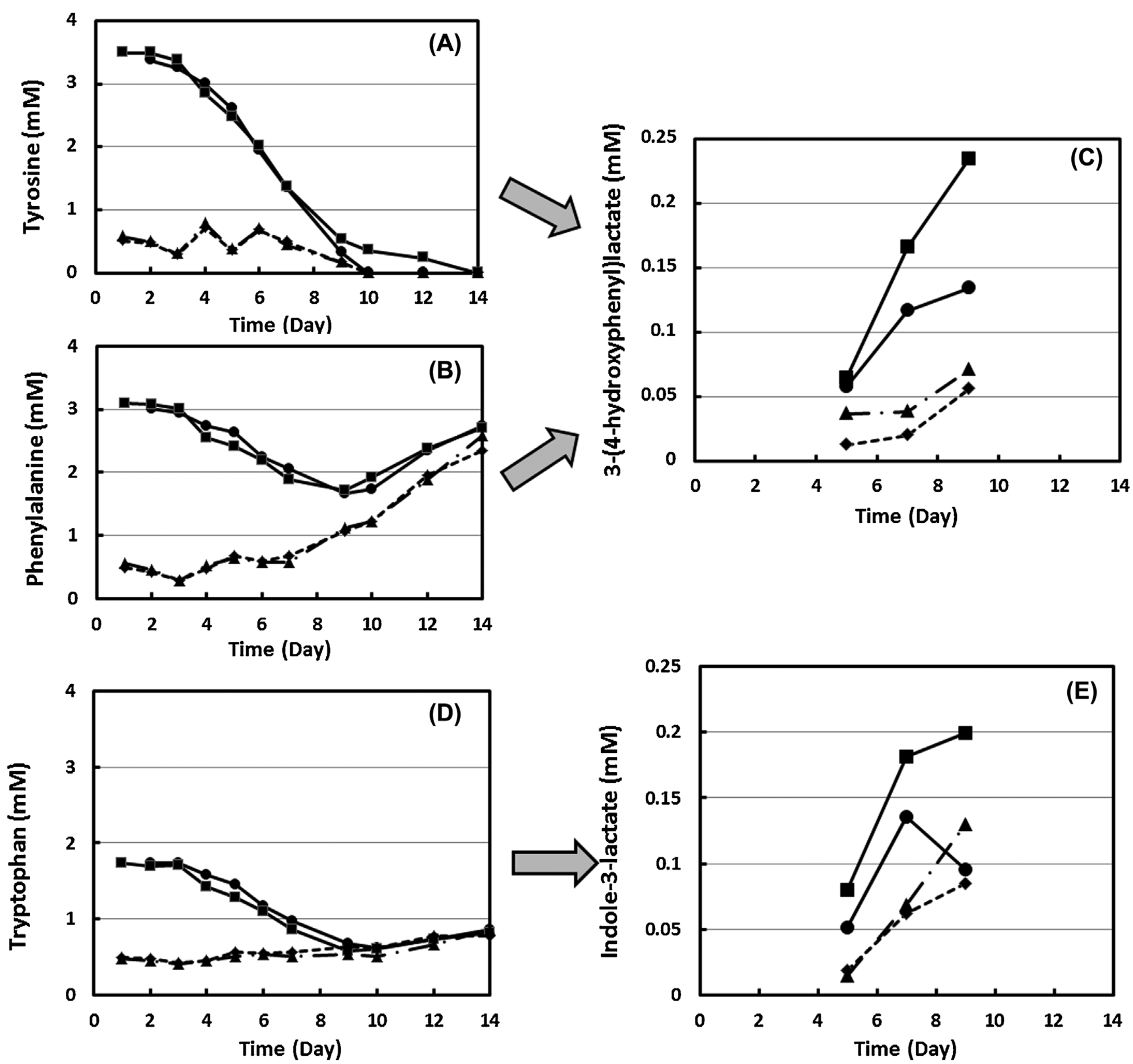

Figure 5. Effect of nutrient limitation on accumulation of the novel growth inhibitors in HiPDOG fed-batch cultures of CHO cell line A. HiPDOG cultures of CHO cells were carried out by maintaining the levels of eight amino acids (Low 8AA) or four amino acids (Low 4AA) in concentration range 0.5–1 mM for first 7 days of the culture. Control HiPDOG cultures (HiPDOG 1 and 2) were carried at normal amino acid levels. Tyrosine (**A**) and phenylalanine (**B**) were controlled at low levels in the Low 8AA and Low 4AA conditions, which led to lower accumulations of 3-(4-hydroxyphenyl)lactate (**C**) in both the cultures as compared to control cultures. Tryptophan (**D**) was controlled at low levels in the Low 8AA and Low 4AA conditions, which resulted in lower levels of indole 3-lactate (**E**) in both conditions as compared to control cultures. Low 8AA amino acids: tyr, phe, met, trp, leu, ser, met, thr; Low 4AA: tyr, phe, met, trp. Low 8AA (▲), Low 4AA (◆), HiPDOG 1 (■), and HiPDOG 2 (•).

Control of Carbon Source to Limit the Production of Novel Growth Inhibitors in Fed-Batch Cultures

The effect of reducing certain culture amino acid concentrations on biosynthesis of the inhibitors was explored. Three conditions were tested: lower levels of eight amino acids including tryptophan, phenylalanine, tyrosine, methionine, glycine, serine, threonine, and leucine (Low 8AA), lower levels of four amino acids including tryptophan, phenylalanine, tyrosine, and methionine (Low 4AA), or a condition with typical amino acid levels (control; two replicates—HiPDOG1 and HiPDOG2). Exponentially growing cells (cell line A) were seeded at 1×10^6 cells/mL in bioreactors with these conditions. For all three conditions, the HiPDOG strategy was operational between Days 0 and 7 of the culture in order to limit lactate from accumulating beyond growth inhibitory levels. In the low amino acid conditions, the concentrations of the above-mentioned four or eight amino acids were maintained between 0.5 and 1 mM for the first 7 days of the culture after which they were adjusted to the levels closer to the respective amino acid concentrations as measured at that time point (Day 7) in the control HiPDOG conditions. Post-day 7, all the three conditions were treated similarly until Day 12.

The concentrations of the amino acids were successfully maintained between 0.5 and 1 mM in both Low 4AA and Low 8AA conditions until Day 7 of the fed-batch cultures (Fig. S3). Such limitation of amino acid levels in the two conditions resulted in lower levels of accumulation of the newly identified metabolites (Figs. 4 and 5). Isovalerate, formate, 3-(4-hydroxyphenyl)lactate, and indole-3-lactate were specifically profiled. Isovalerate and formate are byproducts of leucine, serine, glycine, and threonine, which are controlled at low levels only in Low 8AA condition. These amino acids are not controlled at low levels in the Low 4AA condition. Correspondingly, significantly lower concentrations of isovalerate were only seen in the Low 8AA condition. The levels of isovalerate were higher in the control HiPDOG conditions and the Low 4AA condition (Fig. 4). Formate levels were similar across all the conditions on Day 7 of the culture; however, on a per cell basis, the amount of formate produced was lower in the Low 8AA condition compared to HiPDOG conditions (Fig. 4). The other two inhibitors profiled, 3-(4-hydroxyphenyl)lactate and indole-3-lactate, are byproducts of the amino acids phenylalanine and tryptophan, which are controlled at lower levels in both Low 8AA and Low 4AA conditions. Significantly lower concentrations of these two inhibitors were observed in both Low 8AA and Low 4AA conditions compared the HiDPOG conditions on Day 7 (Fig. 5).

The cells in the Low 8AA and Low 4AA conditions grew better than the control conditions (HiPDOG1 and HiPDOG2), peaking at cell densities of 50×10^6 and 45×10^6 cells/mL, respectively, on Day 9 whereas the cell densities in control conditions peaked around 32×10^6 cells/mL (Fig. 6A). Such an increase in cell growth observed in the late stages of the low amino acid conditions can be explained as an outcome of the reduced inhibitor accumulations in the culture (Figs. 4 and 5). The low amino acid conditions had only slightly higher titer when compared to the control HiPDOG conditions until Day 9, indicating that cells in low amino acid conditions have lower specific productivity (q_P). Post-day 9, the titers in Low 8AA and Low 4AA conditions tapered off to more

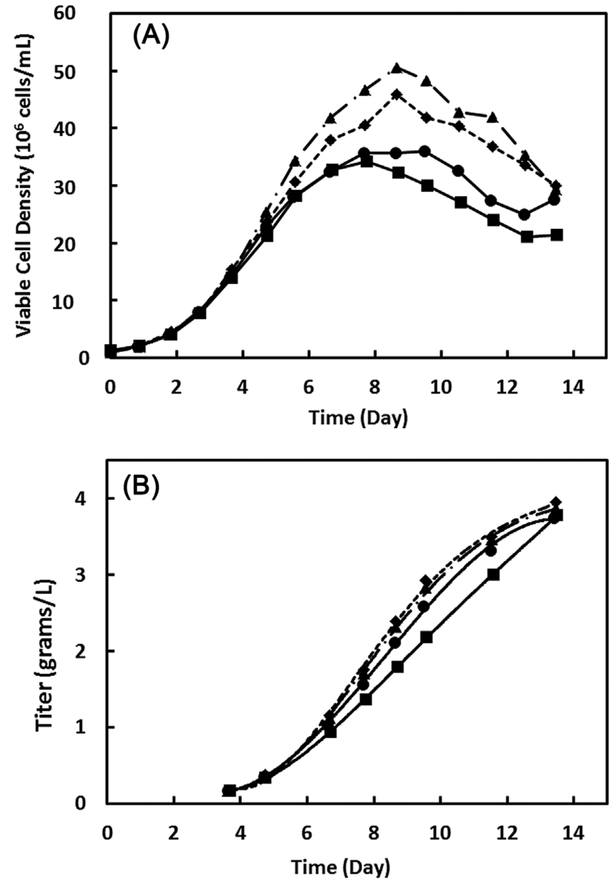

Figure 6. Growth and productivity of CHO cell line A in HiPDOG fed-batch cultures with or without amino acid restriction. HiPDOG cultures of CHO cells were carried out by maintaining the levels of eight amino acids (Low 8AA) or four amino acids (Low 4AA) in concentration range 0.5–1 mM for first 7 days of the culture. Control HiPDOG cultures (HiPDOG 1 and 2) were carried at normal amino acid levels. (**A**) Viable cell density. (**B**) Titer. Low 8AA (▲), Low 4AA (◆), HiPDOG 1 (■), and HiPDOG 2 (●).

closely match the titer levels of control HiPDOG conditions by the end of the culture (Fig. 6B). The post-day 9 reduction in the protein production in the low amino acid conditions was attributed to the near exhaustion of tyrosine levels in the cultures post-day 9 (Fig. S3).

The low amino acid conditions were tested on a second cell line (cell line B) and similar results were obtained (data not shown). Bioreactors with low amino acid conditions accumulated lower levels of inhibitors and reached significantly higher peak cells densities. The higher peak cell densities in low amino acid conditions yielded higher titers when compared to control. This could be explained by reduced accumulation of the novel inhibitors as observed in the low amino acid condition (Fig. S4).

Development of a Generic Process with a Pre-Determined Feeding Schedule to Control the Levels of Amino Acids within the Low Concentration Range

In the earlier experiments, the control of amino acids at low concentrations was accomplished through daily measurement of culture amino acid levels and subsequent supplementation of amino acids individually (using concentrated stock solutions) based on the instantaneous consumption rates such that the amino acid levels at the next sampling point would be near 0.5 mM. For large-scale cGMP manufacturing, such a process would not be convenient and alternative approaches would be required. Therefore, a generic and adaptable strategy for maintaining the residual amino acid levels within the desired low concentration range using a pre-determined feeding schedule was necessary. Since the HiPDOG control strategy uses nutrient feed (containing glucose) to deliver glucose on demand, one approach would be to utilize the HiPDOG strategy to deliver the right amount of amino acids on a daily basis such that amino acid levels always remain within the desired concentration range.

Plotting the amount of feed delivered during the period when HiPDOG strategy is operational against the integral viable cell concentration (IVCC) yields a clear linear correlation (Fig. S5). This suggests that during the HiPDOG phase, the amount of feed delivered up until any given point is proportional to total (or integral) number of viable cells. Furthermore, the specific consumption rates of the amino acids during the growth phase were relatively constant (Fig. S6). Since the feed media added on a per cell basis (through the HiPDOG scheme) and the nutrient uptake rate on a per cell basis are known, the concentration of the amino acids required in the feed medium to supplement the amino acids at the rate they are taken up by cells can be calculated. With such a feed rate, the amino acid levels during the culture could be maintained at the levels closer to those present at the inoculation time point (e.g., ~0.5 mM). This strategy was successfully applied to cell line A (Fig. 7A and B) and cell line B (Fig. 7C and D). In both the cases, amino acid levels were well maintained around the desired low levels (~0.5 mM) for most of the growth phase and gradually increased over the later stages of the culture (Figs. 8 and S7), which should minimize any concern of amino acid mis-incorporations in the recombinant protein of interest produced during the culture. The peak cell densities and the titer obtained in the Low 8AA conditions were higher for both cell lines. Although the titer was higher in Low 8AA condition for both the cell lines, the specific productivity (q_P) was observed to be lower. The cause for such lower q_P in Low 8AA conditions is currently being investigated. In a separate experiment, product quality attributes for material produced by cell line B in Low 8AA and control HiPDOG conditions

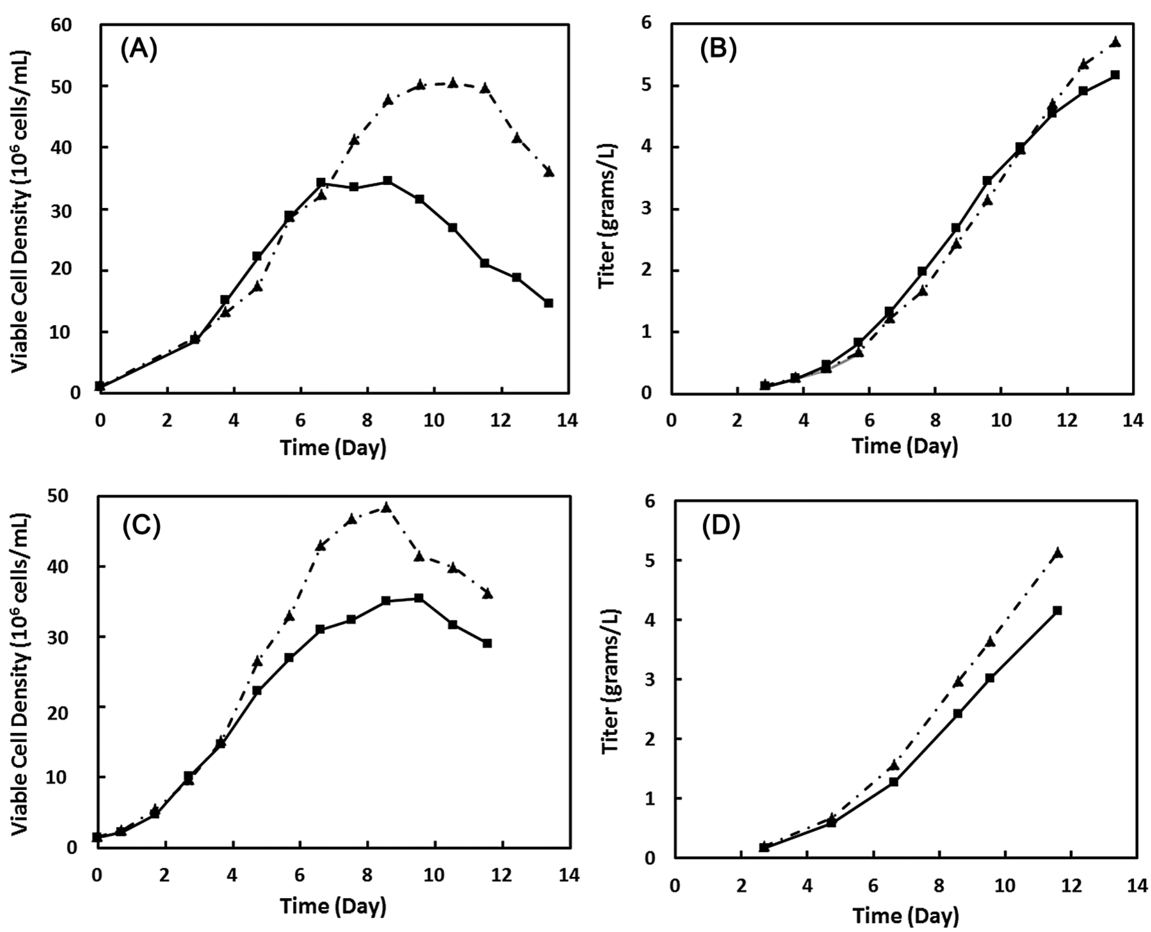

Figure 7. Growth and productivity of CHO cell lines A and B in amino acid restricted HiPDOG fed-batch cultures employing pre-defined feeding strategy for controlling levels of eight amino acids in low concentration range. Viable cell densities (**A**) and culture titer (**B**) of cell line A cultivated in the Low 8AA (▲) or control HiPDOG (■) conditions. Viable cell densities (**C**) and culture titer (**D**) of cell line B cultivated in the Low 8AA (▲) or control HiPDOG (■) conditions.

were assessed (data not shown). The levels of high molecular mass species (HMMS), low molecular mass species (LMMS), and glycosylation attributes across the two conditions were comparable (values within assay variability). Small differences in the levels of acidic ($\Delta \sim 3\%$) and basic ($\Delta \sim 4\%$) species were observed between the two conditions.

Discussion

Proliferating cells consume large amounts of nutrients and divert a significant portion of them to lactate and ammonia production (DeNicola and Cantley, 2015; Glacken, 1988; Young, 2013). Such accumulation of lactate and ammonia is inhibitory to cell growth and productivity, and can also affect recombinant protein product quality (Cruz et al., 2000; Hansen and Emborg, 1994; Kurano et al., 1990; Lao and Toth, 1997; Yang and Butler, 2000). At first glance, the cells might be seen to be exhibiting "suboptimal metabolism," but, it has been argued that such a state is essential for cells to maintain high growth rates (Vander Heiden et al., 2009).

The HiPDOG strategy reported by Gagnon et al. (2011) successfully limits lactate production without significantly slowing cell growth. Furthermore, development of the glutamine synthatase expression system has completely relieved the glutamine dependence of cultured CHO cells and significantly reduced the levels of ammonia accumulation typically seen in fed-batch cultures. Fed-batch cultures employing both the above-mentioned strategies benefit from reduced lactate and ammonia production yielding higher peak cell densities (Figs. 1 and S1). In HiPDOG culture, although ammonia accumulates initially to about 6 mM by Day 4, at which point it might have some inhibitory effect on the growth, the concentration drops to below inhibitory levels by Day 7 ($\sim$1 mM). However, cell growth stops even at such reduced accumulations of lactate and ammonia, indicating a second tier of growth inhibitors potentially coming into play in these cultures when cell densities are sufficiently high late in culture.

Over the last two decades, significant efforts have been spent on understanding glucose metabolism of cultured mammalian cells. However, amino acid metabolism is not well understood. As amino acids are mainly used for protein synthesis, it is imperative to supplement them in large quantities in cell culture medium. There have been reports of amino acid mis-incorporation in proteins due to depletion of specific amino acids in culture (Khetan et al., 2010; Wen et al., 2009). Interestingly, the other extreme, excessively high levels of nutrients in culture can also be suboptimal (see review, Wellen and Thompson, 2010). In the current work, for the first time, we demonstrate that under conditions of low lactate and ammonia, there are additional metabolites produced that negatively affect cell growth, and that these inhibitors are, to an extent, the byproducts from the overflow of amino acid metabolism (Table I). Uptake rates of amino acids higher than those required for protein synthesis can result in channeling of the excess carbon toward byproduct synthesis. Reducing the uptake rates, by limiting the residual medium concentrations of amino acids, can restrict byproduct formation which can translate into higher cell densities and productivities (Figs. 4–6).

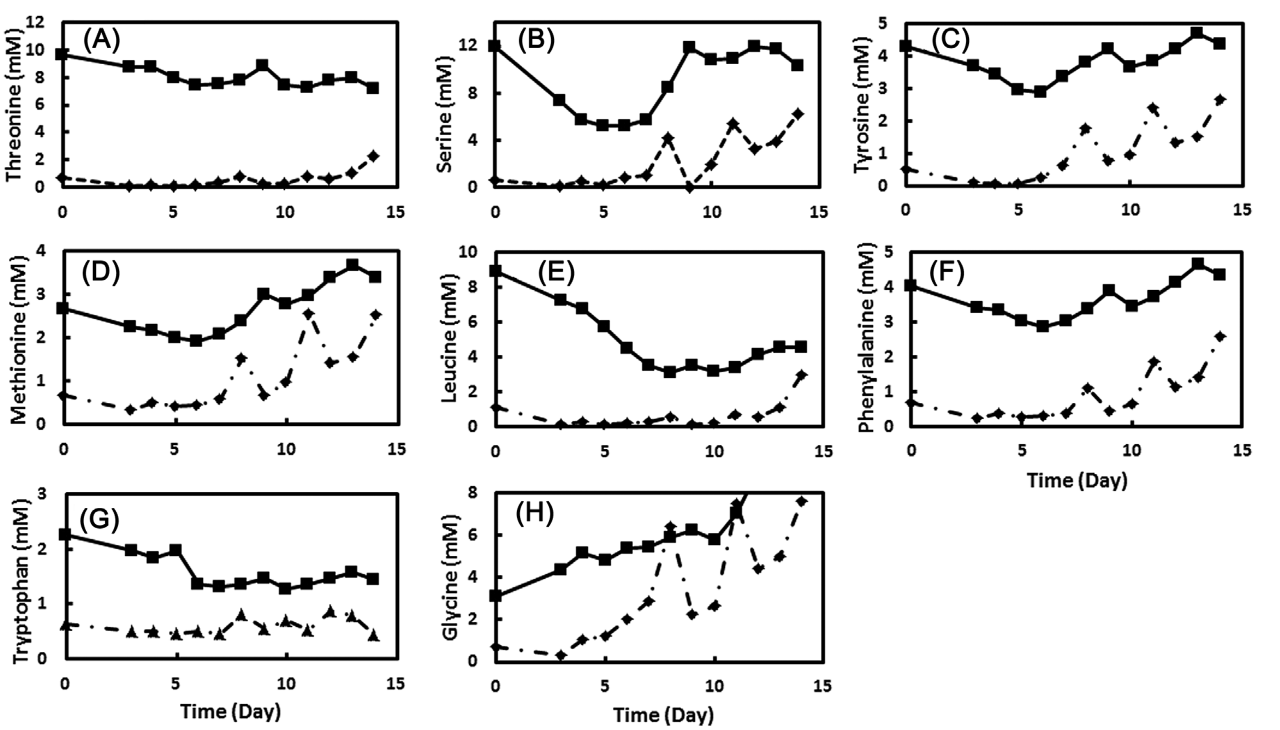

Figure 8. Residual amino acids levels in amino acid restricted and control HiPDOG fed-batch cultures of CHO cell line A. Time course concentrations profiles of threonine (**A**), serine (**B**), tyrosine (**C**), methionine (**D**), leucine (**E**), phenylalanine (**F**), tryptophan (**G**), or glycine (**H**) across the Low 8AA (▲), or control HiPDOG (■) cultures.

The identification of additional cellular byproducts was possible due to availability of metabolomics technologies. Metabolomics allows for the survey of a large number of the metabolites present within the cell or in the extracellular milieu at any given time point of a culture. It is being used on a more routine basis to study the changes in the metabolic phenotypes observed in different types of cell culture processes (Dietmair et al., 2012; Mohmad-Saberi et al., 2013; Sellick et al., 2011; Selvarasu et al., 2012). In the current study, the novel inhibitors were identified as compounds accumulating in the spent medium over the course of the culture. However, not all the accumulating compounds have growth inhibitory effects as a number of compounds were identified that had no negative effect when tested as part of the add-back experiments (data not shown). For those which had an effect on cell growth, the concentrations at which the effect was noticed varied from micromolar to sub-millimolar, to, in some cases, the millimolar range (Fig. 3). Not all compounds accumulate in cultures to the levels at which they have an independent inhibitory effect. In the present study, it was determined that the concentrations of the inhibitors as measured on Day 7 in a HiPDOG culture act synergistically to significantly suppress cell growth (Fig. 3).

Control of inhibitor production can be attained in various ways. As demonstrated in this study, a process can be designed purely on the basis of cell-specific amino acid consumption rates (Fig. 7). By supplementing the amino acids at the rate at which they are consumed, the concentrations can be exquisitely controlled in the low concentration range (Figs. 8 and S7). Alternatively, online or frequent offline measurement techniques can be employed to continuously monitor the concentrations of amino acids. Use of Raman spectroscopy and near-infrared spectroscopy (NIRS) to measure the culture levels of amino acids has been already demonstrated (Riley et al., 2001; Whelan et al., 2012). Such monitoring of amino acid levels might allow for feedback control supplementation whenever the levels drop below the desired set point. In addition to these methods, inhibitors can be flushed out in a semi-continuous or continuous manner from the culture by operating the process in a perfusion mode (with cell retention) (Clincke et al., 2013; Zhang et al., 2015).

In conclusion, we report for the first time the identification of novel growth inhibitors produced by CHO cells in fed-batch cultures. Supplementation of nutrients, specifically amino acids, at high levels in culture medium seems to be the main cause for their production. The potential mechanisms by which these compounds suppress growth are yet to be understood. Furthermore, the inhibitors identified and controlled as part of this study are only a fraction of a larger set that accumulate and inhibit cell growth in fed-batch cultures of CHO cells. Hence, identification of the near complete set of inhibitors and understanding their mechanism of action, along with enhanced knowledge on cell metabolism linked to the inhibitor biosynthesis, should allow development of robust and highly optimized fed-batch processes with volumetric productivities rivaling those obtained from “sustainable” perfusion bioreactor systems.

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